Figure 6: Evolutionary Lineage (b.splosive -> enki.1 -> monad(1)) & Autopoietic Self-Repair Architecture

AST Self-Introspection, 8-Class Failure Autopsy (RC_01-RC_08), LOO-Falsified Residual Bridges, Hippocampal STM Shard, and 4-Tier Dreaming

1. Four-Generation Architectural Lineage: From Reactive Bitboard Kernel to Self-Constructed Autopoietic Mind

GEN 1: b.splosive (v0.2)

SCP & Bicameral Bitboards

* 3.8x BPE SCP & D1-D4 Guard Ladder
* 19 Rust Apps, ArcBitboard10 & MitM

GEN 2: K.MOSH (s)&(d) (v0.4)

Huxley Valve & Conscious Self

* A'<->B' Monologue (Mistral-NeMo-8B)
* Top-Down D4xS10 Group Conjugation

GEN 3: enki.1 (INOS v0.6)

25 LTP Shards & 5 AGI Circuits

* 11,881 Certified Capsules (25 Shards)
* MCTS World Model & AST Self-Audit

GEN 4: monad(1) (INOS v0.7)

Autopoietic Self-Repair Kernel

* RC_01-RC_08 Autopsy & LOO Bridges
* Hippocampal STM & 4-Tier Dreaming

Stage 1: Forward Monadic Solve

AST Critique: 4 Bottlenecks Fixed

2. Organ 1 (Forward Solve) & Organ 2 (Failure Autopsy)

MANDATORY IF Delta_train > 0

Stage 1A-1D: Monadic Forward Solver Cascade (Monad1Kernel.solve_arc_task)

* 1A: O(1) Learned Repair Ops (13 closures) * 1B: 7-Family CanvasDocument Vector Tools
* 1C: Counterfactual MCTS Rollouts (H-descent) * 1D: Bicameral A<->B + Conscious Observer
* **Selects Best Near-Miss Operator f* = argmin_f H(f(X), Y) whenever Delta_train > 0**

Stage 2: 8-Class Quantitative Root-Cause Failure Taxonomy (RC_01 - RC_08)

* **RC_01 (Color Bijection):** Dim match = 1.0 & Foreground IoU >= 0.92, unmapped palette
* **RC_02 (Panel Boolean Reduction):** Separator walls split input into target-sized subpanels
* **RC_03 (Dimension Scaling / Crop):** Non-identity Kronecker scale, block mode, or BBox crop
* **RC_04 (Local 3x3 CA Patch):** Dim match = 1.0 & cell acc >= 78%, local cavity/border delta
* **RC_05 (Colinear Ray Bridge):** >= 75% of residual cells align on H/V marker sightlines
* **RC_06 (MCTS Horizon Shortfall):** Hamiltonian energy H < 0.42 halted 1-2 steps early
* **RC_07 (Test Shift Overfit):** Delta_train = 0 achieved, but test shifted palette or D4 frame
* **RC_08 (Missing DSL Primitive):** Triggers orthogonal all-bridge sweep + DSL spec emission

-> Logs structured FailureAutopsyReport to data/monad1_arc_autopsy_ledger.json

3. Organ 3: Residual Bridge Synthesizer & Epistemic Ratchet

LOO & Delta_train = 0 GATED

Stage 3: Prescribed + Orthogonal Residual Bridge Synthesis (f_repair = b o f*)

* **Family A (Palette Homomorphism):** Direct color map, S10 rank map, or 2-color swap
* **Family B (Panel Boolean/Quorum):** 12 reductions (XOR, AND, OR, NOR, Majority, Unique)
* **Family C (Dimension Morphisms):** Kronecker s_y x s_x, Fractal G(x)G, Block Mode, Crop
* **Family D (LOO-Validated 3x3 CA):** Learns 3x3 table on N-1 pairs; falsifies on N-th pair
* **Family E (Colinear Ray Bridges):** Horizontal/vertical same-color or delta-color sightline fill
* **Family F (Deep Near-Miss MCTS):** 2-step rollout from f*(X_m) + Tabular OOF Bridges

Admitted iff forall m, f_repair(X_m) == Y_m (plus 100% LOO holdout for local CA rules)

Cumulative Epistemic Ratchet (O(1) Semigroup Promotion + SHARD_25 Injection)

1. Prepends verified monad1_repair_* closure to learned_operators[0] for O(1) priority
2. Persists witness training pairs to disk for deterministic rehydration across sessions
3. Hot-injects Kandel LTP synapse into SHARD_25_ARC3_VISUAL_PUZZLE_OOS in < 1 ms

Ledger: 1,894 Cycles | 11,375 Evals | 10,569 Stage-1 + 16 Rescues | 13 Promoted Ops

Online DOM Transitions (prev, a, next)

Task Completion -> Adaptive Consolidation

4. Hippocampal Short-Term Memory (STM_SHARD_00) & 4-Tier Adaptive Dream Consolidation

1,316 STM->LTM TRANSITIONS | 491 DREAM CYCLES

Monad1ShortTermMemoryShard (Online Trial-and-Error Physics Learner)

* **HTML5 Canvas RGB Retina:** 16-color exact quantization (4,096/4,096 pixels) on arcprize.org
* **O(1) Episodic Graph:** 100% pruning of proven-inert actions (S_{t+1} == S_t) & 2-step loops
* **Empirical Causal Discovery:** Infers avatar locus, walls, pushable crates, chasm bridges, clickable switches ACTION6(r,c), phase gates, keys/locks, portals, & Baba rule sockets
* **Imagined World-Model Lookahead:** Multi-step BFS strictly over learned STMCausalMechanics
* **Loop-Erased STM->LTM Reduction:** Compresses trace into shortest path pi*_min on task exit

Epistemic Surprise Gate:

S_surprise = min(1, 0.14*max(0, N_laws - 3) + 0.45*max(0, rho_comp - 1) + 0.16*N_inert + 0.55*I[~won])

Online Eval: 8 / 8 (100%) Live Browser Levels + D1-D5 Unseen Curriculum (100% Solved)

Monad1DreamConsolidationEngine (4-Tier Adaptive Reflection & Sleep Hierarchy)

Tier 0 - FAST_SYNAPTIC_TAG (0.0 ms, S < 0.20, 168 cycles): Zero-friction solve (rho_comp = 1.0).
Commits loop-free STM->LTM tag directly into SHARD_25; skips heavy dreaming overhead.

Tier 1 - PONTIFICATE (~0.05 ms, 0.20 <= S < 0.40 or stall, 68 cycles): Bounded micro-reflection
over active STM graph; prunes A<->B 2-cycles & emits deductive syllogisms (SYLLOGISM_COUPLED_BRIDGE).

Tier 2 - DAYDREAM (~0.5 ms, 0.40 <= S < 0.70 or brief idle, 165 cycles): Wakeful associative replay
of latest episode under S10 palette-permuted 1-corridor counterfactual perturbation.

Tier 3 - DEEP_SLOW_WAVE (~1.4 ms, S >= 0.70 or >= 6 eps, 132 cycles): SWS 16-episode replay
(gamma_SWS = 0.94 downscaling) + Multi-Mechanic REM Counterfactual Dream Synthesis into SHARD_25.